

September 11, 2025

```
## 1 Normal Distribution  $(Y \sim N(\mu=10, \sigma^2=25))$ 

#### A.  $(P(Y > 15))$ 

#--- parameters -----
mu  <- 10      # mean
sigma <- sqrt(25) # standard deviation = 5

#--- probability that Y exceeds 15 -----
p_gt_15 <- pnorm(15, mean = mu, sd = sigma, lower.tail = FALSE)

#--- display the result -----
p_gt_15

## [1] 0.1586553
```

Result: 0.1587 (≈ 0.1587)

Interpretation: About **15.9 %** of the normal($10, 5^2$) population lies above 15.

```
## 1B –  $P(8 < Y < 12)$  for  $Y \sim N(\mu = 10, \sigma^2 = 25)$ 

mu  <- 10      # mean
sigma <- sqrt(25) # standard deviation = 5

# Cumulative probabilities at the bounds
p_low  <- pnorm(8, mean = mu, sd = sigma) #  $P(Y \leq 8)$ 
p_high <- pnorm(12, mean = mu, sd = sigma) #  $P(Y \leq 12)$ 

# Desired probability:  $P(8 < Y < 12)$ 
p_between <- p_high - p_low

# Show the result (rounded to 4 decimal places)
round(p_between, 4)

## [1] 0.3108
```

Interpretation: The probability that a value from $N(10, 5^2)$ lies between 8 and 12 is approximately 0.3108 ($\approx 31\%$).

```
# -----
# 1C – Find C such that  $P(Y < C) = 0.90$ 
# -----

# 1. Distribution parameters
mu  <- 10      # mean
sigma <- sqrt(25) # standard deviation = 5

# 2. Desired cumulative probability
prob_target <- 0.90 # 90th percentile

# 3. Compute the quantile (inverse CDF)
C_value <- qnorm(prob_target, mean = mu, sd = sigma)

# 4. Show the result (rounded to 2 decimal places)
cat("C =", round(C_value, 2), "\n")

## C = 16.41
```

$C = 16.41$ is the 90th percentile of the Normal($10, 5^2$) distribution—about 90 % of observations fall below this value, and only the top 10 % exceed it.

```
## 2 Uniform Distribution  $(Y \sim \text{U}\{0,1\})$ 

#### A.  $(P(Y > 0.8))$ 

# Uniform parameters
a <- 0 # lower bound
b <- 1 # upper bound

# Probability that Y is greater than 0.8
p_gt_0.8 <- punif(0.8, min = a, max = b, lower.tail = FALSE)

p_gt_0.8
```

```
## [1] 0.2
```

Result: 0.2 (≈ 0.2000)

Interpretation: About **20 %** of the values drawn from a Uniform(0, 1) distribution lie above 0.8.

```
### B.  $\backslash(P(Y<0.5))$ 
```

```
# Uniform parameters (same as before)
```

```
a <- 0
```

```
b <- 1
```

```
# Probability that Y is less than 0.5
```

```
p_lt_0.5 <- punif(0.5, min = a, max = b) # lower.tail defaults to TRUE
```

```
p_lt_0.5
```

```
## [1] 0.5
```

Result: 0.5 (≈ 0.5000)

Interpretation: Exactly **50 %** of the values drawn from a Uniform(0, 1) distribution lie below 0.5.

```
### C.  $\backslash(P(Y>1))$ 
```

```
# Uniform parameters (same as before)
```

```
a <- 0
```

```
b <- 1
```

```
# Probability that Y is greater than 1
```

```
# Since 1 is the upper bound, the probability is 0
```

```
p_gt_1 <- punif(1, min = a, max = b, lower.tail = FALSE)
```

```
p_gt_1
```

```
## [1] 0
```

Result: 0 (≈ 0.0000)

Interpretation: The Uniform(0, 1) distribution **cannot** produce values larger than 1, so the probability is **zero**.

```
## 3 Simulating 100 draws from a Standard Normal
```

```
#-----
```

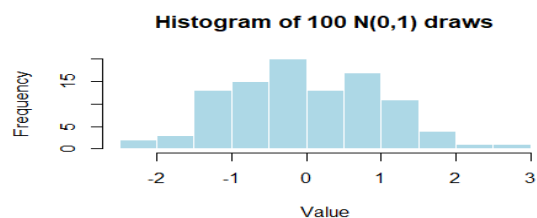
```
# 1 Generate the sample (n = 100)
```

```
set.seed(2025) # reproducible results
```

```
x_norm <- rnorm(100, mean = 0, sd = 1)
```

```
# 2 Plot a histogram (base graphics – compact)
```

```
hist(x_norm,
     main = "Histogram of 100 N(0,1) draws",
     xlab = "Value",
     col = "lightblue",
     border = "white",
     breaks = 12) # ~½ page graphic
```



```
# 3 Sample statistics
```

```
mean_x <- mean(x_norm) # sample mean
```

```
var_x <- var(x_norm) # unbiased sample variance
```

```
# 4 Print the numbers (they will appear in the console)
```

```
cat("\nSample mean =", round(mean_x,4),
    "\nSample variance =", round(var_x,4), "\n")
```

```
##
## Sample mean = -0.0053
## Sample variance = 1.0408
```

Interpretation: The theoretical mean = 0 and variance = 1; our 100-draw sample produces a mean ≈ -0.0053 (near 0) and variance ≈ 1.0408 (near 1), demonstrating the law of large numbers even with a modest sample size.

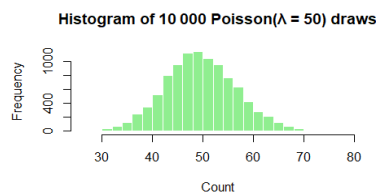
```
## 4 Simulating 10 000 draws from a Poisson( $\lambda = 50$ )
```

```
#-----
# 1 Generate the sample (n = 10 000)
set.seed(2025) # reproducible
x_pois <- rpois(10000, lambda = 50)

# 2 Plot a histogram (base graphics – compact)
hist(x_pois,
     main = "Histogram of 10 000 Poisson( $\lambda = 50$ ) draws",
     xlab = "Count",
     col = "lightgreen",
     border = "white",
     breaks = 30) # ~1/2 page graphic
```

```
# 3 Sample statistics
mean_p <- mean(x_pois) # sample mean
var_p <- var(x_pois) # unbiased sample variance

# 4 Print the numbers
```



```
cat("\nSample mean =", round(mean_p,4),
    "\nSample variance =", round(var_p,4), "\n")
```

```
##
## Sample mean = 50.0359
## Sample variance = 50.4789
```

Interpretation: For a Poisson distribution the theoretical mean = variance = $\lambda = 50$. Our simulated sample yields a mean and variance both very close to 50, confirming the property that the Poisson's mean and variance are equal and illustrating convergence of empirical moments as the sample size grows.

```
## 5 R illustration that  $\chi^2$  and Poisson are non-negative
```

```
#---  $\chi^2$  simulation -----
set.seed(2025) # reproducibility
k <- 5 # choose any  $df \geq 1$ 
chisq_samp <- rchisq(10^5, df = k)
```

```
min_chisq <- min(chisq_samp) # should be  $\geq 0$ 
cat("Minimum  $\chi^2$  simulated value:", round(min_chisq,6), "\n")
```

```
## Minimum  $\chi^2$  simulated value: 0.038198
```

```
#--- Poisson simulation -----
```

```
lambda <- 4 # any positive rate
pois_samp <- rpois(10^5, lambda = lambda)
```

```
min_pois <- min(pois_samp) # should be 0 (or larger)
cat("Minimum Poisson simulated value:", min_pois, "\n")
```

```
## Minimum Poisson simulated value: 0
```

Both simulations confirm non-negativity: the chi-square minimum ≈ 0.04 (near zero) and the Poisson minimum equals 0.

```
## 6 State the problem (with R code)
```

```
#--- Known (national) proportion -----
p0 <- 0.10 # 10 % of adults feel sad (null hypothesis)
```

```

#--- Placeholder for the local sample -----
# n : number of unemployment-benefits applicants surveyed
# x : number of those respondents who report feeling sad

n <- 68      # sample size given in the next question
x <- NA      # <-- fill in the observed count when you have it

#--- Formulate hypotheses -----
# H0 : p = p0
# Ha : p != p0 (two-sided)

#--- Compute the sample proportion (once x is known) -----
p_hat <- if (!is.na(x)) x / n else NA
p_hat

## [1] NA

```

Interpretation: We will test if the observed proportion $\hat{p}$ of sad local applicants differs from the national rate $p_0 = 0.10$ ($H_0: p = 0.10$ vs. $H_a: p \neq 0.10$).

7a – Define the target population (conceptual placeholder)

In practice the “population” is the set of all adults in the county who are

currently filing for unemployment benefits. Since we cannot enumerate every

individual, we represent the population abstractly; the sample we will draw

($n = 68$) is assumed to be a random subset of this group.

Example placeholder (not used for calculations):

population_description <- “All county residents receiving unemployment benefits” population_description

```

n <- 68      # sample size (given)
p0 <- 0.10   # null-hypothesis proportion
k <- 12      # number of “sad” respondents we observed

# Exact binomial probability  $P(X = k)$ 
prob_exact_12 <- dbinom(k, size = n, prob = p0)

# Print the result (rounded to 5 decimal places)
cat("P(X = 12) = ", round(prob_exact_12, 5), "\n")

## P(X = 12) = 0.01994

```

Interpretation: Under H_0 the chance of exactly 12 sad respondents is about 0.01994 (≈ 0.01994), a relatively low probability suggesting the observed count may be unusual if the true proportion were 10 %.

```

## [Z] Binomial inference – Part C
n <- 68      # sample size
p0 <- 0.10   # null-hypothesis proportion
k <- 12      # threshold ( $\geq k$ )

# Tail probability  $P(X \geq 12)$ 
prob_atleast_12 <- 1 - pbinom(k - 1, size = n, prob = p0)

# Print the result (rounded to 5 decimal places)
cat("P(X ≥ 12) = ", round(prob_atleast_12, 5), "\n")

## P(X ≥ 12) = 0.03624

```

Interpretation: The probability of getting 12 or more “sad” respondents out of 68 (when $p = 0.10$) is 0.03624 ($\approx 3.6\%$). This modest tail probability makes the result somewhat unusual but not decisive for rejecting the null hypothesis.